

● TMCS

TRAFFIC MONITORING CAMERA SYSTEM

A Safer, Smarter Highway Starts With What Your Cameras Already See

TMCS turns every highway camera into an automated incident detector — stopped vehicles, wrong-way driving, pedestrians, debris and more — confirmed in real time, backed by court-grade evidence, and built to slot straight into your control room and ATMS.

🔍 NHAH ATMS Policy 2023 — aligned

🔗 Multi-vendor ONVIF cameras

🛡️ Evidence-grade & auditable



7 Incident Types

Detected automatically on every camera



Multi-Vendor

ONVIF-based — any PTZ camera brand



Evidence-Grade

Tamper-evident, court-ready forensic trail



Phased Rollout

Pilot-to-corridor, signed off stage by stage

Built for continuous highway operation — from a single-camera pilot to a full corridor rollout, commissioned in verifiable phases.

What TMCS Delivers

TMCS (Traffic Monitoring Camera System) converts live feeds from highway IP/PTZ cameras into **reliable, explainable incident events** — stopped and wrong-way vehicles, pedestrians and animals on the carriageway, road debris, and building congestion — together with tamper-evident forensic evidence, without asking a control-room operator to watch every screen at once.

It is the surveillance and detection layer beneath a highway Advanced Traffic Management System (ATMS): TMCS does the continuous watching and the first-pass judgement call, and hands the downstream system a structured, versioned, audit-traceable event — not a raw video stream to be re-interpreted by someone else, later, under pressure.



Automated Detection

7 canonical highway incident types, evaluated continuously across every registered camera, day and night.



Evidentiary Integrity

Every confirmed event is backed by a tamper-evident, write-once evidence package — image, metadata, and the exact model/rule versions that fired it.



Integration-Ready

Built as a distinct layer from ATMS/VMS/VIDES from day one, so downstream systems consume clean events instead of re-deriving them from video.

The challenge TMCS solves

A highway control room cannot scale headcount with camera count. As a corridor grows from a handful of junctions to hundreds of kilometres of coverage, manual monitoring stops being a plan and becomes a gap — incidents get noticed late, evidence is inconsistent, and there is no reliable record of what a camera actually saw when it matters most. TMCS closes that gap at the camera, automatically, before it ever reaches a screen.

TMCS is vendor-agnostic at the device layer (ONVIF-based) — it sits under an existing highway camera estate and does not require replacing installed cameras to add automated detection to them.

Alignment with NHAI's ATMS Policy (2023)

The National Highways Authority of India's ATMS Standards & Specifications, announced **17 October 2023** (circular 11.53/2023), position the Traffic Monitoring Camera System as a core subsystem of the highway Advanced Traffic Management System — cameras deployed at approximately **1 km intervals** along National Highways, upgraded for **automated detection of accidents and stalled/stopped vehicles**, feeding a control-room layer rather than requiring continuous manual monitoring.

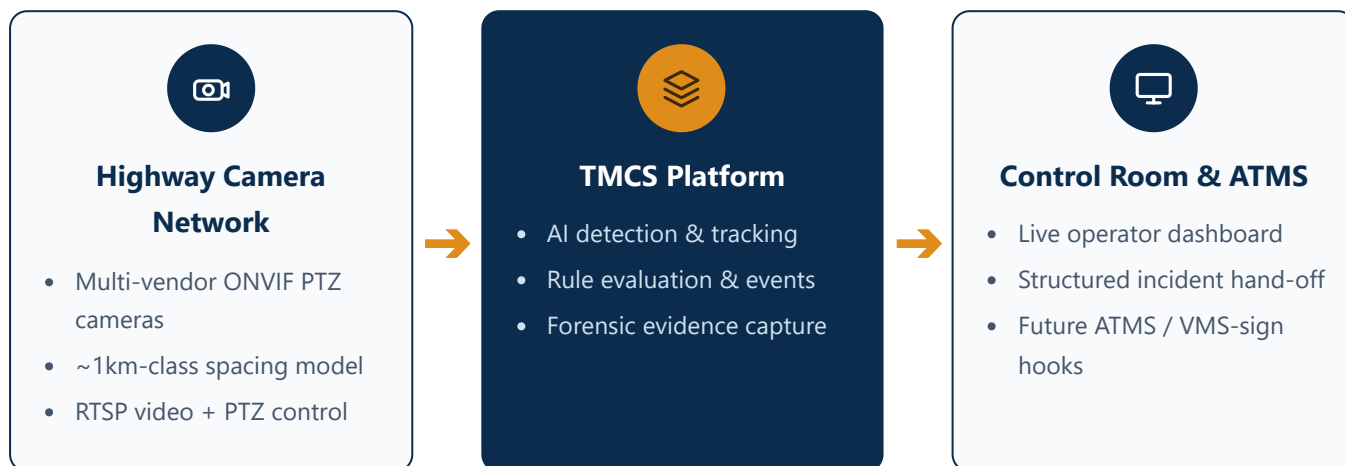
TMCS is architected directly against that functional intent:

- ✓ **Automated incident detection** at every camera location, not manual review of live feeds, matching the policy's shift away from pure human monitoring.
- ✓ **Standards-based device layer** (ONVIF) so an approximately-1km camera deployment model can be built from multi-vendor PTZ cameras rather than a single proprietary product.
- ✓ **TMCS as a distinct subsystem** beneath ATMS — the same separation the policy draws between the camera/detection layer and the traffic-management decision layer, so TMCS can be commissioned and validated independently of the wider ATMS rollout.
- ✓ **An integration layer built for hand-off** — confirmed events, evidence, and vehicle classification are designed to be consumed by an ATMS / control-room / VMS-message-sign layer, in line with the policy's control-room-centric operating model.

Compliance note: NHAI's published policy and its Functional & Technical Specifications (Volume II) govern the project. TMCS is designed in line with their functional intent; final compliance for any specific award is always validated against that project's tender, BOQ, and technical specification annexures — this document is a solution overview, not a substitute for a formal compliance matrix.

How TMCS Fits Into the Highway Ecosystem

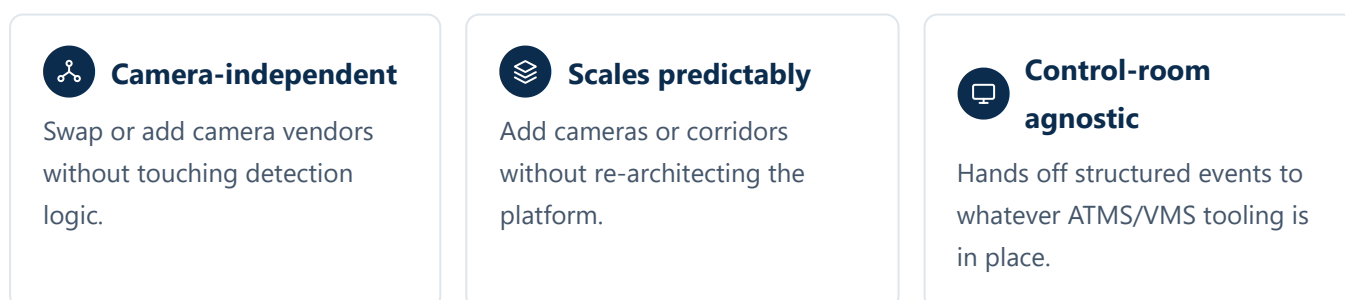
At the highest level, TMCS sits between the camera network and the people and systems who act on what those cameras see — turning raw footage into decisions, not just pictures.



Camera feeds enter as video; only clean, structured, evidence-backed events leave toward the control room.

Why the separation matters

Keeping these three zones distinct is a deliberate design choice, not an accident of how the system happened to be built. It means the camera estate can grow, camera vendors can change, and the control room's tools can evolve — each independently — without the other two needing to change in step. A highway authority is rarely replacing its entire camera network, its detection platform, and its control room software on the same day; TMCS is built so it doesn't have to.



A Layered Platform, Not a Monolith

Each responsibility inside TMCS is owned by exactly one layer. This is what keeps the platform reliable as camera count, vendors, and rule complexity all grow — a change in one layer cannot silently break another.



Why it matters commercially: a layered platform is what makes a phased, low-risk rollout possible in the first place — each layer can be commissioned and signed off on its own (see the phased delivery model later in this document), instead of the whole system needing to work perfectly before any of it can go live.

From Camera Feed to Confirmed Incident

TMCS never treats a raw detection as an event, and never treats an event as an operator alert, until every stage below has actually confirmed it — this is what keeps false alarms low without hiding real incidents:



A candidate must satisfy temporal and spatial conditions — sustained confidence over time, correct lane/direction, inside the relevant detection zone — before it is confirmed as an event.

Preset-aware, not just camera-aware

Highway PTZ cameras move — to a wide patrol view, to a close-up on an incident, and back. TMCS treats each PTZ preset as its own validated scene: detection zones, lane geometry, and calibration all belong to the specific preset, and analytics only run once the camera has verifiably reached and stabilized on that preset. A camera drifting mid-move, or an operator panning away to inspect something, never produces a false reading against the wrong scene geometry.

7 Canonical Highway Incident Types

Every incident type below runs continuously, on every camera, with no manual triggering.



HIGHEST PRIORITY

Stopped Vehicle

A vehicle halted on the carriageway or shoulder beyond a configurable dwell time — the leading indicator NHA's policy calls out explicitly, and the single largest cause of secondary collisions on high-speed corridors.

Confirmed via: sustained near-zero displacement inside the lane/shoulder zone, tracked over the dwell window so vehicles briefly slowed in traffic are not falsely flagged.



ENFORCEMENT-ORIENTED

Parked Vehicle

Longer-duration stationary vehicles, distinct from a brief stop — typically in a no-parking shoulder zone — flagged separately so removal/enforcement workflows aren't mixed up with urgent breakdown response.

Confirmed via: an extended stationary threshold beyond the stopped-vehicle window, evaluated in the same zone/lane context.



CRITICAL SAFETY

Wrong-Way Driving

Vehicle movement against the configured lane direction — a leading precursor to head-on collisions and one of the fastest-moving risks a control room needs to know about.

Confirmed via: tracked trajectory compared against the lane's configured direction vector, not a single frame's heading.



SITUATIONAL AWARENESS

Queue / Congestion

Building traffic density or reduced flow that signals an upstream incident or bottleneck — often the earliest visible sign that something has gone wrong further up the corridor.

Confirmed via: tracked vehicle density and flow rate within the zone, sustained over a configurable window.

Incident Types, Continued



CRITICAL SAFETY

Pedestrian on Highway

Person presence in a carriageway zone where pedestrians should never be — one of the most time-sensitive alerts TMCS raises, given closing speeds on a national highway.

Confirmed via: person-class detection sustained inside a carriageway/no-pedestrian zone, tracked to rule out a momentary false trigger.



RURAL & FORESTED CORRIDORS

Animal on Road

Livestock or wildlife incursion onto the carriageway — a common and under-reported hazard on highway stretches that run through agricultural or forested land.

Confirmed via: animal-class detection inside the carriageway zone, tracked and classified separately from vehicle and pedestrian classes.



PREVENTIVE

Debris Detected

Fallen cargo, tyre fragments, or other road-surface hazards that can cause a secondary accident long after the vehicle that dropped them has left the scene.

Confirmed via: static-object detection on the road surface, persisting across frames rather than a passing shadow or reflection.

Vehicle & object classification

Detections are classified against an 11-class taxonomy aligned to VIDES vehicle categories, so every event and every piece of evidence is labelled consistently:

Car

Motorcycle

Truck

Bus

Mini Bus

Auto Rickshaw

Tractor

Bicycle

Pedestrian

Animal

Road Debris

Additional incident types and classes can be enabled per project without architectural change — the rule engine and taxonomy are catalog-driven, not hard-coded per deployment.

Forensic Evidence, Built to Be Trusted

A confirmed incident is only useful to a control room, an auditor, or a court if the evidence behind it can be trusted. TMCS treats evidence as a first-class, non-negotiable output:



Write-once, never overwritten

Confirmed-event evidence (snapshot, metadata) is written once and is never silently replaced or edited by later processing.



Full version traceability

Every event records the exact AI model version, rule version, and scene/calibration configuration in force at the moment it fired.



Clean live view, rich evidence view

Operators see a clean, low-latency live stream; the system independently renders bounding boxes, track ID, classification, and timestamp onto the immutable evidence image.



Full audit trail

Every state transition — detection → candidate → confirmed → resolved — is timestamped, sourced, and reasoned, for after-the-fact review.

Why this is worth insisting on

A stopped-vehicle alert with no defensible evidence behind it is a liability, not an asset — for enforcement, for insurance claims, and for public trust in the system. TMCS is built so that every alert an operator acts on already has its evidence trail attached, before anyone has to go looking for it.

Core Platform Capabilities



Multi-vendor camera support

ONVIF-standard discovery, capability negotiation, and PTZ control — new camera vendors are added as adapters, without touching detection or rule logic.



Multi-camera GPU scheduling

A single GPU-resident inference engine cycles across all registered cameras, avoiding per-camera VRAM duplication and keeping capacity predictable as the estate grows.



Live operator dashboard

Real-time incident feed, camera fleet health, PTZ joystick control, preset management, and zone/lane editing in one console.



Automatic PTZ return

After manual PTZ inspection, cameras automatically return to their default surveillance preset on a configurable timeout — no operator has to remember to reset the view.



Resilient device handling

Unsupported or partially-compliant camera features degrade gracefully instead of failing the whole device; PTZ commands carry timeouts, retries, and self-healing fallback behaviour.



Role-based access & audit

Fine-grained permissions for viewing, operating (PTZ/stream control), and configuring (cameras/zones/users), with a full security audit log.

Engineering acceptance gates used internally include sustained ingestion throughput of **25+ frames per second per camera** and **p95 inference latency under 45 ms**, validated per deployment through benchmarking — capacity for any given project is confirmed against its actual camera count, hardware, and scene complexity rather than assumed.

Security, Privacy & Operational Trust

- ✓ Camera credentials, API tokens, and ONVIF service access are never stored in source code or written to logs.
- ✓ Role-based access control with least-privilege defaults for viewing, operating, and configuring the system.
- ✓ Every administrative and control action (user/role changes, camera edits, PTZ commands) is captured in a security audit trail.
- ✓ Configuration changes (presets, zones, rules, models) follow a draft → validate → approve → publish lifecycle — nothing goes live silently, and historical configuration is never overwritten.
- ✓ The system is fully testable without physical cameras (simulator / recorded footage / shadow mode), so new rules and models are proven safe before touching a live highway.

Deployment & Phased Rollout

TMCS is delivered in controlled phases, each independently verifiable, rather than as one large untested cut-over:

Phase	Delivers
1 — Camera Foundation	Camera registry, ONVIF discovery, device health, RTSP verification
2 — PTZ & Presets	PTZ control, preset capture/validation, command audit trail
3 — Scene Configuration	Preset-scoped zones, lane geometry, direction, calibration
4 — Detection	Vehicle / person / animal / object detection and tracking
5 — Event Engine	All 7 canonical incident rules, candidate confirmation
6 — Evidence	Snapshot, pre/post buffers, hashing, metadata packaging
7 — Operations	Event review, commissioning tooling, replay, health, audit
8 — Integration	ATMS, VMS wall, patrol, emergency dispatch, external APIs

Deployment can be scoped to a single pilot camera through to a full corridor rollout; each phase is commissioned and signed off before the next is switched on for live operation.

Why TMCS



Explainable, not a black box

Every confirmed event traces to the exact model, rule, and configuration version that produced it — defensible in audit and in court.



Vendor-independent

ONVIF-based device layer means the camera estate isn't locked to one manufacturer, and new vendors are onboarded as adapters.



Built for the policy, not retrofitted to it

Designed against NHAI's ATMS functional model from the outset — automated detection, control-room hand-off, 1km-class deployment.



Operationally honest

Degrades gracefully instead of failing silently; never conflates "command accepted" with "action completed," so operators can trust the state they see.

Next Steps

- ✓ **Site & camera assessment** — review the target corridor's existing camera estate, spacing, and network reach against ONVIF/PTZ compatibility.
- ✓ **Pilot deployment** — commission TMCS on a small number of cameras to validate detection accuracy and evidence quality on real site conditions before wider rollout.
- ✓ **Compliance mapping** — walk the project's specific tender/BOQ and NHAI technical specification annexures against the TMCS capability set, item by item.
- ✓ **Phased rollout & commissioning** — scale camera-by-camera or corridor-by-corridor, with sign-off at each phase.

Contact

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This document is a solution overview prepared for discussion purposes. Figures describing engineering targets are internal acceptance criteria, not contractual performance guarantees; final scope, SLAs, and compliance commitments are defined in the executed project agreement and the applicable NHAI tender documentation.